MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION, October 2025
SCHEME FOR VALUATION

Part A – Multiple Choice Questions (10 × 1 = 10 marks)

Q.No	Key	CO	Marks
1	c	CO1	1
2	c	CO1	1
3	d	CO1	1
4	a	CO2	1
5	c	CO2	1
6	a	CO2	1
7	c	CO2	1
8	b	CO3	1
9	a	CO3	1
10	c	CO3	1

Part B – Short Answer Questions (Any 4 × 5 = 20 marks)

Each question carries **5 marks**.

Q. No	Expected Points	Marking Scheme	CO
11	**Explain IOT concept, highlighting its key components and providing examples of its applications.** Define IoT, 3–4 key components (sensors, connectivity, data processing, user interface), and 2–3 examples (smart home, wearables, agriculture). **Typical answer:** The Internet of Things (IoT) refers to a network of physical objects—devices, vehicles, sensors, appliances—connected to the Internet that collect and exchange data. Key components:	Definition 1, Components 2, Examples 2	CO1

	1. Sensors/Devices: Detect physical parameters such as temperature, light, or motion. 2. Connectivity: Wi-Fi, Bluetooth, LoRa, or cellular networks to transfer data. 3. Data Processing: Microcontrollers or cloud platforms analyze the received data. 4. User Interface: Mobile apps or dashboards that present results to users. Examples: Smart home lighting, fitness trackers, and agricultural soil-moisture monitors.		
12	**Mention some common Arduino APIs used for programming ESP32 and their functions.** Mention 4–5 APIs (pinMode, digitalWrite, analogRead, Serial.begin, delay, WiFi.begin, server.on). Explain the purpose of each briefly. **Typical answer:** 1. pinMode(pin, mode) – Sets a pin as INPUT or OUTPUT. 2. digitalWrite(pin, value) – Writes HIGH/LOW to a digital pin. 3. digitalRead(pin) – Reads the digital state of a pin. 4. analogRead(pin) – Reads an analog voltage input. 5. Serial.begin(baudrate) – Starts serial communication. 6. delay(ms) – Pauses execution for given milliseconds. 7. WiFi.begin(ssid, password) – Connects ESP32 to Wi-Fi. Each API simplifies hardware control and communication tasks in IoT projects.	Any 5 APIs × 1 mark	CO1
13	**What is the use of Serial.print() function in ESP32? Explain with a simple example** Explain Serial.print() and its role in debugging. Give example Serial.print("Value: "); Serial.println(val);. **Typical answer:** Serial.print() sends data from the ESP32 to the Serial Monitor, useful for debugging or displaying values. Example: int sensor = 25; Serial.begin(9600); Serial.print("Sensor value: "); Serial.println(sensor); This prints “Sensor value: 25” on the Serial Monitor.	Explanation 2, Example 3	CO2

	Serial.print() prints data on one line, whereas Serial.println() moves to the next line.		
14	**Write the syntax of analogRead() function and explain its use** Syntax: analogRead(pin);Use: to read analog voltage (0–4095 in ESP32). Explain example usage. **Typical answer:** Syntax: analogRead(pin); It reads analog voltage (0–3.3 V) from the specified pin and returns a value (0–4095 for ESP32). Example: int val = analogRead(34); Serial.println(val); Used for reading sensors such as LDR, temperature sensors, and potentiometers where output voltage varies continuously.	Syntax 1, Explanation 2, Example 2	CO2
15	**Discuss the advantages and limitations of using gas sensors in indoor air quality monitoring.** Advantages: early detection, low cost, automation. Limitations: calibration, cross-sensitivity, humidity effect. **Typical answer:** Advantages: 1. Detect hazardous gases quickly. 2. Compact and inexpensive. 3. Easy to interface with microcontrollers. 4. Provide real-time alerts. Limitations: 1. Require frequent calibration for accuracy. 2. Affected by temperature and humidity. 3. Limited gas selectivity – cross-sensitivity to other gases. 4. Sensor life shortens with heavy exposure.	Advantages 2, Limitations 3	CO2
16	**List the advantages of connecting ESP32 to a local web server.** Advantages of local web server: no internet required, faster access, secure data, local monitoring, cost-effective. **Typical answer:** 1. No Internet dependency – works within local Wi-Fi range. 2. Fast data access and response time.	Any 5 points × 1 mark	CO3

	3. Secure and private – data stays inside the network. 4. Low operating cost – no cloud subscription needed. 5. Enables real-time control of IoT devices through browsers or phones.		

Part C – Essay Questions (Any 2 × 10 = 20 marks)

Each carries **10 marks**.

Q. No	Key Points Expected	Marking Scheme	CO
17	**Discuss the relevance of internet of things (IOT) in modern society, including its benefits and challenges.** Define IoT, importance in society, 4–5 benefits (automation, efficiency, health, environment), 3–4 challenges (privacy, cost, standardization). **Typical answer:** The Internet of Things (IoT) has revolutionized how humans interact with technology, merging the physical and digital worlds. It connects everyday objects to the Internet, allowing them to sense, communicate, and act intelligently. Benefits: IoT offers automation and efficiency by enabling smart homes, intelligent transportation, precision agriculture, and health-monitoring systems. It optimizes energy usage, reduces human effort, and improves productivity in industries. For instance, smart grids balance power supply and demand automatically. Challenges: Despite its advantages, IoT faces issues such as data security, privacy violations, lack of standardization, and power management for low-energy devices. Additionally, initial setup costs and network reliability are concerns in developing regions.	Introduction 2, Benefits 3, Challenges 3, Conclusion 2	CO1
18	**Apply the concept of servo motors and DC gear motors to design a simple robotic arm mechanism.** Explain servo + DC motor basics, circuit diagram / logic, Arduino control, conceptual design of robotic arm, real-life applications. **Typical answer:** A robotic arm combines multiple actuators such as servo motors and DC gear motors to perform precise and coordinated movement. Servos are used for controlled	Concept 2, Design 4, Code explanation 2, Application 2	CO2

	angular motion in joints, while DC gear motors provide torque for base rotation or lifting actions. In a basic design, the base uses a DC gear motor for rotation, and shoulder and elbow joints use servos to bend the arm. A gripper at the end, powered by a micro-servo, holds or releases objects. The system is controlled by an Arduino or ESP32 through PWM (Pulse Width Modulation) signals. This design is widely applied in pick-and-place operations, educational robotics, and prosthetic systems. Using servos for precision and DC motors for strength offers a balance between flexibility and power in IoT-based automation.		

19	**Write a detailed Arduino program for ESP32 to read a push-button state and turn on an LED only when the button is pressed. Include explanations of key functions used.**
	Complete Arduino sketch using digitalRead and digitalWrite; correct syntax (if (digitalRead(button)==LOW) digitalWrite(LED,HIGH); else digitalWrite(LED,LOW);). Explain functions.
	
	Typical answer:
	An ESP32 program to control an LED through a push-button is given below:
	
	`#define BUTTON 4`
	`#define LED 13`
	
	`void setup() {`
	`  pinMode(BUTTON, INPUT_PULLUP);`
	`  pinMode(LED, OUTPUT);`
	`  Serial.begin(9600);`
	`}`
	
	`void loop() {`
	`  int state = digitalRead(BUTTON);`
	`  if (state == LOW) {`
	`  digitalWrite(LED, HIGH);`
	`  Serial.println("Button Pressed");`
	`  } else {`
	`  digitalWrite(LED, LOW);`
	`  }`
	`}`
		Code 6, Explanation 4	CO2

	The pinMode() function sets the pin as input or output. The digitalRead() function reads the push-button status (LOW when pressed), while digitalWrite() controls the LED state. Serial.begin() initializes serial communication, and Serial.println() prints messages for debugging. This simple control mechanism demonstrates the fundamentals of human–machine interaction in IoT systems.		
20	**Describe the advantages and limitations of ESP32 Wi-Fi networking compared to other IoT connectivity options.** Compare ESP32 Wi-Fi with ZigBee, LoRa, Bluetooth; cover speed, range, power, cost, applications; discuss pros & cons. **Typical answer:** The ESP32 is a dual-core microcontroller with built-in Wi-Fi and Bluetooth, widely used in IoT systems for its speed and flexibility. It enables fast data transfer up to 150 Mbps, making it ideal for smart homes, monitoring dashboards, and automation projects requiring real-time connectivity. Compared with other technologies, ESP32 Wi-Fi provides higher data rates than Bluetooth Low Energy (BLE) or ZigBee, though it consumes more power. LoRa, in contrast, covers several kilometers but offers very low data speed, making it suitable for long-range environmental sensing. The advantages of ESP32 Wi-Fi include its integrated wireless hardware, easy Internet connectivity through a full TCP/IP stack, and strong developer support. However, it consumes more power and has a limited indoor range of around 30–50 m. Network congestion in the 2.4 GHz band may also affect performance. Overall, ESP32 Wi-Fi is ideal for medium-range IoT applications that demand high-speed, reliable data transfer, whereas BLE, ZigBee, or LoRa are better suited for low-power or long-distance use.	Comparison table/points 6, Evaluation 4	CO3